

Solutions to MathPhys II colloquium 2 & exam 1

Colloquium, problem 1. Spherical reservoir R_1, D_C, R_2, D_R

Initial condition: $c(r, 0) = \Theta(R_1 - r)$, boundary condition: $C(R_2, t) = 0$.

a) We choose spherical coordinates (r, θ, φ) , centered at the center of the reservoir. Because of the symmetry (isotropy): $c = c(r, t)$. Because the diffusion constant is different in the two domains, let's define:

$$c_n(r, t) = \begin{cases} c_n^{(C)}(r, t); & 0 \leq r \leq R_1 \\ c_n^{(R)}(r, t); & R_1 \leq r \leq R_2 \end{cases}.$$

We look for the solution using the separation of variables:

$$c_n(r, t) = \rho_n(r) \tau_n(t).$$

We're solving:

$$D_C \frac{\nabla_r^2 \rho_n^{(C)}}{\rho_n^{(C)}} = \frac{\dot{\tau}_n^{(C)}}{\tau_n^{(C)}}; \quad 0 \leq r \leq R_1,$$

$$D_R \frac{\nabla_r^2 \rho_n^{(R)}}{\rho_n^{(R)}} = \frac{\dot{\tau}_n^{(R)}(t)}{\tau_n^{(R)}(t)}; \quad R_1 \leq r \leq R_2.$$

Since the function will have to be continuous (and the current as well) at $R_1 = R_2$, this will only be possible for all times if the functions in the two domains have equal time-dependence ($\tau_n^{(C)} = \tau_n^{(R)} =: \tau_n$). So:

$$\frac{\dot{\tau}_n}{\tau_n} = -\kappa_n,$$

$$\nabla_r^2 \rho_n^{(C)} + \frac{\kappa_n}{D_C} \rho_n^{(C)} = 0; \quad 0 \leq r \leq R_1, \tag{1}$$

$$\nabla_r^2 \rho_n^{(R)} + \frac{\kappa_n}{D_R} \rho_n^{(R)} = 0; \quad R_1 \leq r \leq R_2. \tag{2}$$

This is solved by:

$$\tau_n(t) = e^{-\kappa_n t},$$

$$\rho_n^{(C)}(r) = j_0 \left(\sqrt{\frac{\kappa_n}{D_C}} r \right) = \frac{\sin(k_n^{(C)} r)}{k_n^{(C)} r},$$

$$\rho_n^{(R)}(r) = B_n j_0 \left(\sqrt{\frac{\kappa_n}{D_R}} r \right) + C_n y_0 \left(\sqrt{\frac{\kappa_n}{D_R}} r \right) = B_n \frac{\sin(k_n^{(R)} r)}{k_n^{(R)} r} - C_n \frac{\cos(k_n^{(R)} r)}{k_n^{(R)} r},$$

with the short-hand notation $k_n^{(C)} := \sqrt{\frac{\kappa_n}{D_C}}$ and $k_n^{(R)} := \sqrt{\frac{\kappa_n}{D_R}}$. In the domain $0 \leq r \leq R_1$ we have taken just the spherical Bessel function j_0 , because the spherical Neumann function y_0 diverges at $r = 0$. In the domain $R_1 \leq r \leq R_2$ we need to take both. The equations (1) and

(2) form a Sturm-Liouville problem on $0 \leq r \leq R_2$ and as we know from the lectures, it follows from here that the functions:

$$\rho_n(r) = \begin{cases} \rho_n^{(C)}(r); & 0 \leq r \leq R_1 \\ \rho_n^{(R)}(r); & R_1 \leq r \leq R_2 \end{cases},$$

form a complete orthogonal basis on the full interval $0 \leq r \leq R_2$.

Boundary conditions:

The function $c(r, t)$ has to satisfy the following boundary conditions:

Continuity at $r = R_1$: $c^{(C)}(R_1, t) = c^{(R)}(R_1, t)$:

$$\frac{\sin(k_n^{(C)} R_1)}{k_n^{(C)} R_1} e^{-\kappa_n t} = B_n \frac{\sin(k_n^{(R)} R_1)}{k_n^{(R)} R_1} e^{-\kappa_n t} - C_n \frac{\cos(k_n^{(R)} R_1)}{k_n^{(R)} R_1} e^{-\kappa_n t}, \quad (3)$$

Continuity of (radial) current $\vec{j}_r = -\hat{e}_r D \frac{\partial c}{\partial r}$ at $r = R_1$: $\vec{j}_r^{(C)}(R_1, t) = \vec{j}_r^{(R)}(R_1, t)$:

$$D_C \frac{k_n^{(C)} R_1 \cos(k_n^{(C)} R_1) - \sin(k_n^{(C)} R_1)}{k_n^{(C)} R_1^2} e^{-\kappa_n t} = B_n D_R \frac{k_n^{(R)} R_1 \cos(k_n^{(R)} R_1) - \sin(k_n^{(R)} R_1)}{k_n^{(R)} R_1^2} e^{-\kappa_n t} + C_n D_R \frac{k_n^{(R)} R_1 \sin(k_n^{(R)} R_1) + \cos(k_n^{(R)} R_1)}{k_n^{(R)} R_1^2} e^{-\kappa_n t} \quad (4)$$

where the time dependent parts canceled out, because we have set $\tau_n^{(C)} = \tau_n^{(R)} = \tau_n = e^{-\kappa_n t}$.

Boundary condition at $r = R_2$: $c^{(R)}(R_2, t) = 0$:

$$B_n \sin(k_n^{(R)} R_2) e^{-\kappa_n t} - C_n \cos(k_n^{(R)} R_2) e^{-\kappa_n t} = 0. \quad (5)$$

Equations (3) and (4) are a set of linear equations using which we can find B_n and C_n . It was ok if you just wrote down the set of equations in the colloquium without solving it. The equation (5) is a transcendental equation $\tan(k_n^{(R)} R_2) = \frac{C_n}{B_n}$, with solutions $k_n^{(R)} R_2 = \xi_n$, $n = 1, 2, \dots$ (of course finding the exact solution was also not required). This gives us the quantization condition $\kappa_n = D_R \frac{\xi_n^2}{R_2^2}$. For simplicity of notation, we will keep writing formulas using B_n , C_n , $k_n^{(C)}$, $k_n^{(R)}$ and keep in mind that B_n , C_n are fixed with the above linear equations and $k_n^{(C)} = \sqrt{\frac{D_R}{D_C} \frac{\xi_n}{R_2}}$, $k_n^{(R)} = \frac{\xi_n}{R_2}$.

Initial condition:

The full solution will be a linear combination:

$$c(r, t) = \sum_n A_n \rho_n(r) e^{-\kappa_n t}.$$

To fix the constants A_n , we need to satisfy the initial condition:

$$A_n = \frac{\langle c(r, 0), \rho_n(r) \rangle}{\langle \rho_n(r), \rho_n(r) \rangle},$$

where we have used orthogonality of $R_n(r)$. Using the formula

$$\int x \sin(kx) dx = \frac{\sin(kx)}{k^2} - \frac{x \cos(kx)}{k} + C, \quad (6)$$

which we could simply copy from the handbook of mathematical functions (or obtain using integration by parts), we have:

$$\begin{aligned} \langle c(r, 0), \rho_n(r) \rangle &= c_0 \int_0^{2\pi} d\varphi \int_{-1}^1 d\theta \int_0^{R_2} \Theta(R_1 - r) \frac{\sin(k_n^{(C)} r)}{k_n^{(C)} r} r^2 dr \\ &= \frac{c_0 4\pi}{k_n^{(C)}} \int_0^{R_1} r \sin(k_n^{(C)} r) dr \\ &= \frac{c_0 4\pi}{(k_n^{(C)})^2} \left(\frac{\sin(k_n^{(C)} R_1)}{k_n^{(C)}} - R_1 \cos(k_n^{(C)} R_1) \right). \end{aligned}$$

Using the formulae:

$$\begin{aligned} \int \sin^2(kx) dx &= \frac{1}{2} x - \frac{1}{4k} \sin(2kx) + C, \\ \int \cos^2(kx) dx &= \frac{1}{2} x + \frac{1}{4k} \sin(2kx) + C, \\ \int \sin(kx) \cos(kx) dx &= \frac{1}{2k} \sin^2(kx) + C, \end{aligned}$$

from the handbook of mathematical functions, we get:

$$\begin{aligned} \langle \rho_n(r), \rho_n(r) \rangle &= 4\pi \int_0^{R_2} \rho_n^2(r) r^2 dr \\ &= \frac{4\pi}{(k_n^{(C)})^2} \int_0^{R_1} \sin^2(k_n^{(C)} r) dr + \\ &\quad + \frac{4\pi}{(k_n^{(R)})^2} \int_{R_1}^{R_2} [B_n^2 \sin^2(k_n^{(R)} r) + C_n^2 \cos^2(k_n^{(R)} r) - 2B_n C_n \sin(k_n^{(R)} r) \cos(k_n^{(R)} r)] dr \\ &= \frac{4\pi}{(k_n^{(C)})^2} \left(\frac{1}{2} R_1 - \frac{1}{4k_n^{(C)}} \sin(2k_n^{(C)} R_1) \right) + \\ &\quad + \frac{4\pi}{(k_n^{(R)})^2} \left(\frac{1}{2} [B_n^2 + C_n^2] [R_2 - R_1] + \frac{1}{4k_n^{(R)}} [C_n^2 - B_n^2] [\sin(2k_n^{(R)} R_2) - \sin(2k_n^{(R)} R_1)] \right) + \\ &\quad + \frac{1}{k_n^{(R)}} B_n C_n [\sin^2(k_n^{(R)} R_2) - \sin^2(k_n^{(R)} R_1)]. \end{aligned}$$

This completes part a).

b) After a long time, the most slowly decaying mode (with the smallest κ) dominates, that is:

$$c(r, t)|_{t \gg 1} \approx A_1 \rho_1(r) e^{-\kappa_1 t}.$$

The flow $I(t)$ of chemical into the surrounding of the reservoir can then be found using $\vec{j} = \vec{j}_r = -\hat{e}_r D \frac{\partial c}{\partial r}$ and $I = \int dS \hat{n} \cdot \vec{j}$. By evaluating the current at $r = R_2$ (using the derivative which we have already computed in (4)), we get

$$I(t) = e^{-\kappa_1 t} 4\pi R_2^2 D_R A_1 \left(B_1 \frac{k_1^{(R)} R_2 \cos(k_1^{(R)} R_2) - \sin(k_1^{(R)} R_2)}{k_1^{(R)} R_2^2} + C_1 \frac{k_1^{(R)} R_2 \sin(k_1^{(R)} R_2) + \cos(k_1^{(R)} R_2)}{k_1^{(R)} R_2^2} \right).$$

Exam, problem 1. Tube in a shape of a section of a circle, R , φ , v_1 , v_2 . As we have seen a couple of times in the exercise classes, since the velocity can only be along the tube (in the z direction), the problem is equivalent to solving the 2D Laplace equation:

$$\nabla_{2D}^2 v_z(r, \vartheta) = 0,$$

in polar coordinates (r, ϑ) with boundary conditions $v_z(r, 0) = v_1$, $v_z(r, \varphi) = 0$, $v_z(R, \vartheta) = v_2$. We've seen in the lectures that the general solution of the Laplace equation in polar coordinates is:

$$v_z(r, \vartheta) = (1 + E \ln r) (F + G\vartheta) + \sum_{m=1}^{\infty} (r^{\lambda_m} + C_m r^{-\lambda_m}) (A_m \sin(\lambda_m \vartheta) + B_m \cos(\lambda_m \vartheta)). \quad (7)$$

We also see immediately that in our case $C_m = 0$, for $\forall m$, because $r^{-\lambda_m}$ diverge at $r = 0$. For the same reason $E = 0$.

The boundary conditions are nonhomogeneous. But we know from the lectures that, since the Laplace equation is linear, we can split such a problem into subproblems, so that in each subproblem we take care of one of the boundary conditions. In the end we can add these solutions to get a solution of the full problem (satisfying all the boundary conditions). So let's split the problem into the following subproblems:

i) Find the function $v_z^{(i)}(r, \vartheta)$ such that: $v_z^{(i)}(r, 0) = v_1$, $v_z^{(i)}(r, \varphi) = 0$. Here we will not be able to satisfy $v_z^{(i)}(R, \vartheta) = 0$, but we don't have to care about it, since what remains at $r = R$, that is $v_z^{(i)}(R, \vartheta)$, we can simply attribute to the next subproblem.

ii) Find $v_z^{(ii)}(r, \vartheta)$ such that: $v_z^{(ii)}(r, 0) = 0$, $v_z^{(ii)}(r, \varphi) = 0$, $v_z^{(ii)}(R, \vartheta) = v_2 - v_z^{(i)}(R, \vartheta)$. In the last BC we have subtracted what remained unsatisfied by the first subproblem.

Now we need to determine the undetermined constants (F , G , A_m , B_m and λ_m) for both subproblems.

Subproblem i):

At $\vartheta = 0$, the following functions appear as a linear combination in the general solution: a constant (multiplied by F) and the functions $\{r^{\lambda_m}\}_m$ (multiplied by A_m). Since the boundary condition $v_z^{(i)}(r, 0) = v_1$ seen as a polynomial is simply a zeroth order polynomial (or a constant), we can satisfy it by setting $F = v_1$. We won't need any of the r^{λ_m} here. Then to satisfy the BC $v_z^{(i)}(r, \varphi) = 0$, we can fix $G = \frac{-v_1}{\varphi}$. At $r = R$ we have $v_z^{(i)}(R, \vartheta) = v_1 - \frac{v_1}{\varphi} \vartheta$ and we attribute

that to the next subproblem.

Subproblem ii):

To satisfy $v_z^{(ii)}(r, 0) = 0$, we set $B_m = 0; \forall m$. To get $v_z^{(ii)}(r, \varphi) = 0$, we need to set $\lambda_m = \frac{m\pi}{\varphi}$. To find A_m , we have to satisfy $v_z^{(i)}(R, \vartheta) = v_2 - v_1 + \frac{v_1}{\varphi}\vartheta$:

$$v_2 - v_1 + \frac{v_1}{\varphi}\vartheta = \sum_{m=1}^{\infty} A_m R^{\frac{m\pi}{\varphi}} \sin\left(\frac{m\pi}{\varphi}\vartheta\right).$$

Multiply on both sides with $\sin(\frac{n\pi}{\varphi}\vartheta)$ and integrate $\int_0^\varphi d\vartheta$. We've seen it many times in the exercise classes (or we could have derived it using simple integrations):

$$\int_0^\varphi \sin\left(\frac{n\pi}{\varphi}\vartheta\right) d\vartheta = \frac{(1 + (-1)^{n+1}) \varphi}{n\pi}$$

and

$$\int_0^\varphi \sin\left(\frac{m\pi}{\varphi}\vartheta\right) \sin\left(\frac{n\pi}{\varphi}\vartheta\right) d\vartheta = \frac{\varphi}{2} \delta_{m,n}.$$

Using (6), we can get:

$$\int_0^\varphi \vartheta \sin\left(\frac{n\pi}{\varphi}\vartheta\right) d\vartheta = \frac{\varphi^2 (-1)^{n+1}}{n\pi}.$$

Thus:

$$\begin{aligned} A_m &= \frac{(v_2 - v_1) \int_0^\varphi \sin\left(\frac{m\pi}{\varphi}\vartheta\right) d\vartheta + \frac{v_1}{\varphi} \int_0^\varphi \vartheta \sin\left(\frac{m\pi}{\varphi}\vartheta\right) d\vartheta}{R^{\frac{m\pi}{\varphi}} \int_0^\varphi \sin^2\left(\frac{m\pi}{\varphi}\vartheta\right) d\vartheta} \\ &= \frac{2}{m\pi R^{\frac{m\pi}{\varphi}}} (-v_1 + (1 + (-1)^{m+1}) v_2). \end{aligned}$$

And we've solved the problem. We could have equally solved it even without the subproblem trick, if we simply took the general solution of the Laplace equation and blindly started satisfying the boundary conditions at $v_z(r, 0) = v_1$. The calculation would lead us through the same procedure.

Colloquium, prob 1., Exam, prob. 2. Diffusion decay equation:

$$\dot{c}(z, t) = D \nabla^2 c(z, t) - \frac{1}{\tau} c(z, t), \quad (8)$$

with initial condition:

$$c(z, 0) = \begin{cases} C_0; & h_1 \leq z \leq h_2 \\ 0; & \text{otherwise} \end{cases}$$

and boundary condition:

$$c(0, t) = 0.$$

a) As the hint suggested, if we plug $w(\vec{x}, t) e^{-\frac{t}{\tau}}$ into (8) we get:

$$\left(w(\vec{x}, t) - \frac{1}{\tau} t \right) \cancel{e^{-\frac{t}{\tau}}} = D \nabla^2 w(\vec{x}, t) \cancel{e^{-\frac{t}{\tau}}} - \frac{1}{\tau} \cancel{e^{-\frac{t}{\tau}}}$$

and indeed the equality holds if $w(\vec{x}, t)$ solves the diffusion equation. Equally, if $G(\vec{x}, t)$ is the Green's function for the diffusion equation:

$$\mathcal{L}_{diff} G(\vec{x}, t) = (\partial_t - D\nabla^2) G(\vec{x}, t) = \delta(\vec{x}, t),$$

then $G(\vec{x}, t)e^{-\frac{t}{\tau}}$ is the Green's function for diffusion decay equation:

$$\begin{aligned} \mathcal{L}_{d-d} \left(G(\vec{x}, t)e^{-\frac{t}{\tau}} \right) &= \left(\partial_t - D\nabla^2 + \frac{1}{\tau} \right) \left(G(\vec{x}, t)e^{-\frac{t}{\tau}} \right) \\ &= \left[\left(\partial_t - \frac{1}{\tau} - D\nabla^2 + \frac{1}{\tau} \right) G(\vec{x}, t) \right] e^{-\frac{t}{\tau}} \\ &= \delta(\vec{x}, t)e^{-\frac{t}{\tau}} \\ &= \delta(\vec{x}, t), \end{aligned}$$

since $e^{-\frac{t}{\tau}} \Big|_{t=0} = 1$. So the Green's function for 1D diffusion decay equation is:

$$G(z, t) = \Theta(t) \left(\frac{1}{4\pi Dt} \right)^{1/2} e^{-\frac{z^2}{4Dt} - \frac{t}{\tau}}.$$

We have chosen the Green's function for the 1D equation since because of translational invariance of the initial condition in x and y directions, the problem is effectively 1D. Now, because for the given problem, the domain is the semi-infinite interval $(-\infty, 0]$ with the BC at 0, we have to find the Green's function satisfying it. As usually, we do that using mirroring (adding a negative δ -source at negative depth):

$$G(z, t; z_0, t_0) = \Theta(t - t_0) \left(\frac{1}{4\pi D(t - t_0)} \right)^{1/2} e^{-\frac{t-t_0}{\tau}} \left(e^{-\frac{(z-z_0)^2}{4D(t-t_0)}} - e^{-\frac{(z+z_0)^2}{4D(t-t_0)}} \right).$$

b) The time-dependence of the concentration profile is obtained using convolution of the initial condition with the Green's function:

$$\begin{aligned} c(z, t) &= \int_{-\infty}^0 c(z_0, 0) G(z, t; z_0, 0) dz_0 \\ &= C_0 \left(\frac{1}{4\pi Dt} \right)^{1/2} e^{-\frac{t}{\tau}} \int_{h_1}^{h_2} \left(e^{-\frac{(z-z_0)^2}{4Dt}} - e^{-\frac{(z+z_0)^2}{4Dt}} \right) dz_0. \end{aligned}$$

Using the definition:

$$\text{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-y^2} dy,$$

together with $\int_a^b dy = \int_0^b dy - \int_0^a dy$ and introducing new variables: $u = \frac{z-z_0}{\sqrt{4Dt}}$, $v = \frac{z+z_0}{\sqrt{4Dt}}$, we find:

$$\begin{aligned} c(z, t) &= \frac{C_0}{2} e^{-\frac{t}{\tau}} \frac{2}{\sqrt{\pi}} \left(\int_{\frac{z-h_2}{\sqrt{4Dt}}}^{\frac{z-h_1}{\sqrt{4Dt}}} e^{-u^2} du + \int_{\frac{z+h_2}{\sqrt{4Dt}}}^{\frac{z+h_1}{\sqrt{4Dt}}} e^{-v^2} dv \right) \\ &= \frac{C_0}{2} e^{-\frac{t}{\tau}} \left(\text{erf} \left(\frac{z-h_1}{\sqrt{4Dt}} \right) + \text{erf} \left(\frac{z+h_1}{\sqrt{4Dt}} \right) - \text{erf} \left(\frac{z-h_2}{\sqrt{4Dt}} \right) - \text{erf} \left(\frac{z+h_2}{\sqrt{4Dt}} \right) \right). \end{aligned}$$

c) In this case we are solving the diffusion equation for a point source. Since there is not translation invariance any more, the problem becomes 3D. Then we can use the 3D Green's function and mirroring:

$$G(\vec{x}, t; \vec{x}_0, t_0) = \Theta(t - t_0) \left(\frac{1}{4\pi D(t - t_0)} \right)^{3/2} \left(e^{-\frac{|(x,y,z)-(x_0,y_0,z_0)|^2}{4D(t-t_0)}} - e^{-\frac{|(x,y,z)-(x_0,y_0,-z_0)|^2}{4D(t-t_0)}} \right).$$

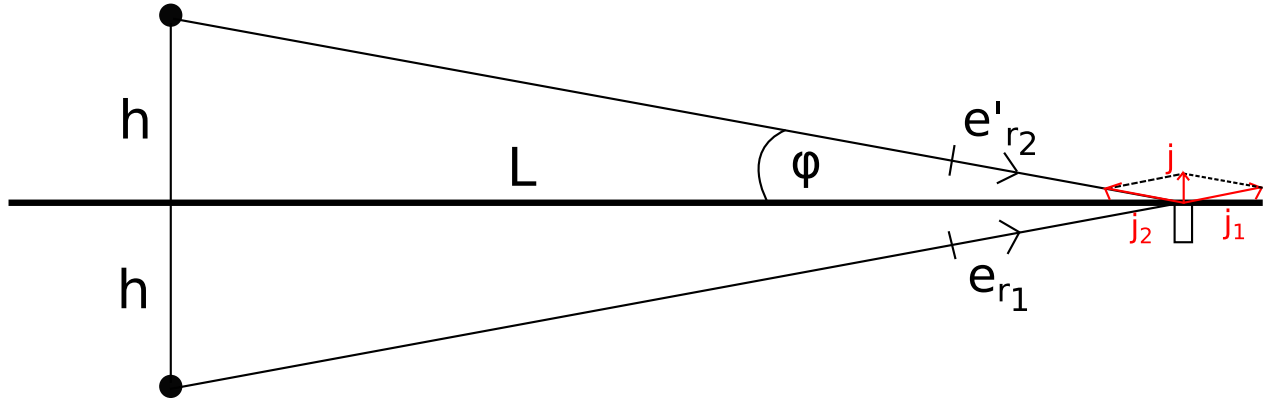
Since we have a point source, the concentration at the is going to be simply

$$\begin{aligned} c(\vec{x}, t) &= \left(\frac{1}{4\pi Dt} \right)^{3/2} \int N \delta(\vec{x}_0 - (0, 0, h)) \left(e^{-\frac{|(x,y,z)-(x_0,y_0,z_0)|^2}{4D(t-t_0)}} - e^{-\frac{|(x,y,z)-(x_0,y_0,-z_0)|^2}{4D(t-t_0)}} \right) d^3 \vec{x}_0 \\ &= N \left(\frac{1}{4\pi Dt} \right)^{3/2} \left(e^{-\frac{|(x,y,z)-(0,0,h)|^2}{4Dt}} - e^{-\frac{|(x,y,z)+(0,0,h)|^2}{4Dt}} \right). \end{aligned} \quad (9)$$

The detector is at the surface and it is reasonable to assume that it is small. The concentration vanishes at the surface. Consequently, the current $\vec{j} = -D\vec{\nabla}c$ is perpendicular to the surface, that is parallel to the detector. Thus we are not going to detect any particles. Alternatively, we can view at (9) as concentration coming from two sources. Each will contribute a current in the radial direction w.r.t the source. For example from the first term:

$$j = \hat{e}_r N \left(\frac{1}{4\pi D(t - t_0)} \right)^{3/2} \frac{2L}{4Dt} e^{-\frac{L^2}{4Dt}}$$

Since one of the terms has a minus sign, the currents are going to add in the direction parallel to the detector (see figure). Again we get that the detected current is going to be zero. Thus, it is not the best idea to place the detector at the surface (remember the Lloyd's mirror problem from the exercise classes).



It was also taken as correct if you did not assume that the detector is small, but rather took it to be of extensive size, for example $S = a \cdot b$. Then the total number of detected particles is (the detector is at $x = L, y = 0, 0 \leq z \leq b$):

$$\begin{aligned} N_{det} &= \int_0^\infty dt \int_0^b dz j_x(L, 0, z, t) \\ &= -aD \int_0^\infty dt \int_0^b dz \frac{\partial}{\partial x} c(\vec{x}, t) \\ &= \frac{aDN2L}{\pi^{3/2}} \int_0^\infty dt \int_0^b dz \left(\frac{1}{4Dt} \right)^{5/2} \left(e^{-\frac{L^2+(z-h)^2}{4Dt}} - e^{-\frac{L^2+(z+h)^2}{4Dt}} \right). \end{aligned}$$

performing first the $\int_0^\infty dt$ integration by introducing a new variable $u = \frac{1}{t}$ and using the formula from the handbook of mathematical formulas (from the section of Laplace transformation):

$$\int_0^\infty \sqrt{t} e^{-pt} dt = \frac{\sqrt{\pi}}{2p^{3/2}},$$

we have:

$$\begin{aligned} N_{det} &= \frac{2aNL}{\pi^{3/2} D^{3/2} 4^{5/2}} \int_0^b dz \int_0^\infty du \sqrt{u} \left(e^{-\frac{L^2+(z-h)^2}{4D}u} - e^{-\frac{L^2+(z+h)^2}{4D}u} \right) \\ &= \frac{aNL}{4\pi} \int_0^b dz \left(\frac{1}{(L^2 + (z-h)^2)^{3/2}} - \frac{1}{(L^2 + (z+h)^2)^{3/2}} \right) \\ &= \frac{aNL}{4\pi} \left(\int_{-h}^{b-h} \frac{1}{(L^2 + u^2)^{3/2}} du - \int_h^{b+h} \frac{1}{(L^2 + u^2)^{3/2}} du \right) \end{aligned}$$

and with the use of

$$\int \frac{dx}{(x^2 + a^2)^{3/2}} = \frac{x}{a^2(x^2 + a^2)^{1/2}},$$

we finally get:

$$N_{det} = -\frac{aNL}{4\pi L^2} \left(\frac{h-b}{\sqrt{\frac{(h-b)^2}{L^2} + 1}} + \frac{h+b}{\sqrt{\frac{(h+b)^2}{L^2} + 1}} - \frac{2h}{\sqrt{\frac{h^2}{L^2} + 1}} \right),$$

which is indeed very very small for $b \ll h \ll L$ (actually even for large detectors and submarine close to the water surface $b < h < L$ - you can check this by inserting numerical values). Note that $h, b < 0$, so that the overall sign is plus.